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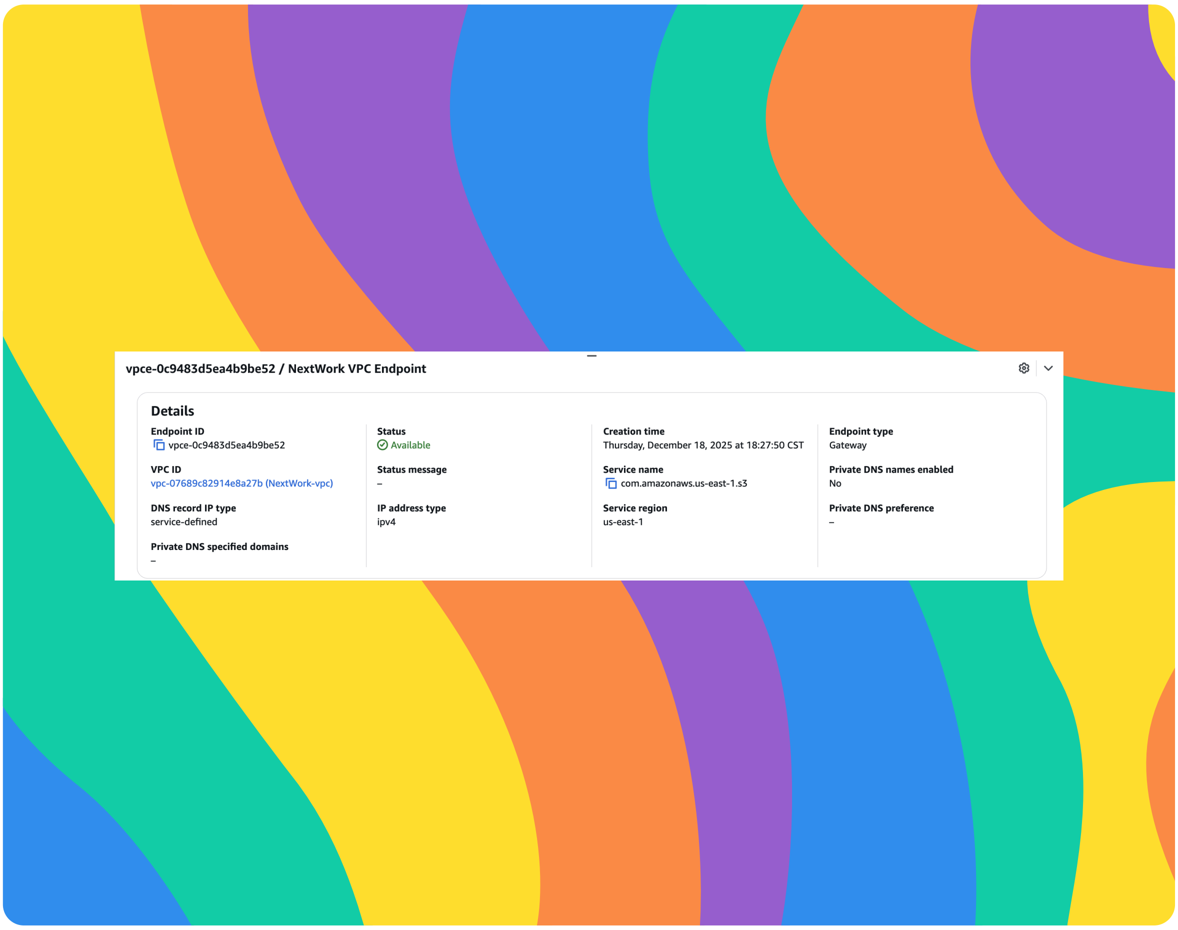
VPC Endpoints

CH Arpita Chowdhury



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated virtual network within AWS where I can launch and control my cloud resources, such as EC2 instances. It is useful because it gives me full control over networking IP addressing, routing, and security allowing mr to securely connect resources, restrict traffic, and keep communication private within the AWS cloud.

How I used Amazon VPC in this project

In today’s project, I used Amazon VPC to create a private network for my EC2 instance and configure an S3 Gateway VPC endpoint, allowing the instance to securely access an S3 bucket without sending traffic over the public internet.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was to not get an error where am suppose to get an error which was accesing the S3 bucket from the EC2 terminal in step 7. I fixed it up later.

This project took me...

This project took me 2 hours.

In the first part of my project...

Step 1 - Architecture set up

In this step, I will create a VPC, launch an EC2 instance and create a S3 bucket because this will set up my architecture.

Step 2 - Connect to EC2 instance

In this step, I will connect my EC2 instance because I want to see what things are like without endpoint connection in place.

Step 3 - Set up access keys

In this step, I will give my EC2 instance access to my AWS enviroment because EC2 needs credential to get access to my AWS services.

Step 4 - Interact with S3 bucket

In this step, I will head back to my EC2 instance and get my EC2 instance to access to my S3 bucket because I want my EC2 insace to connect with my S3 bucket.

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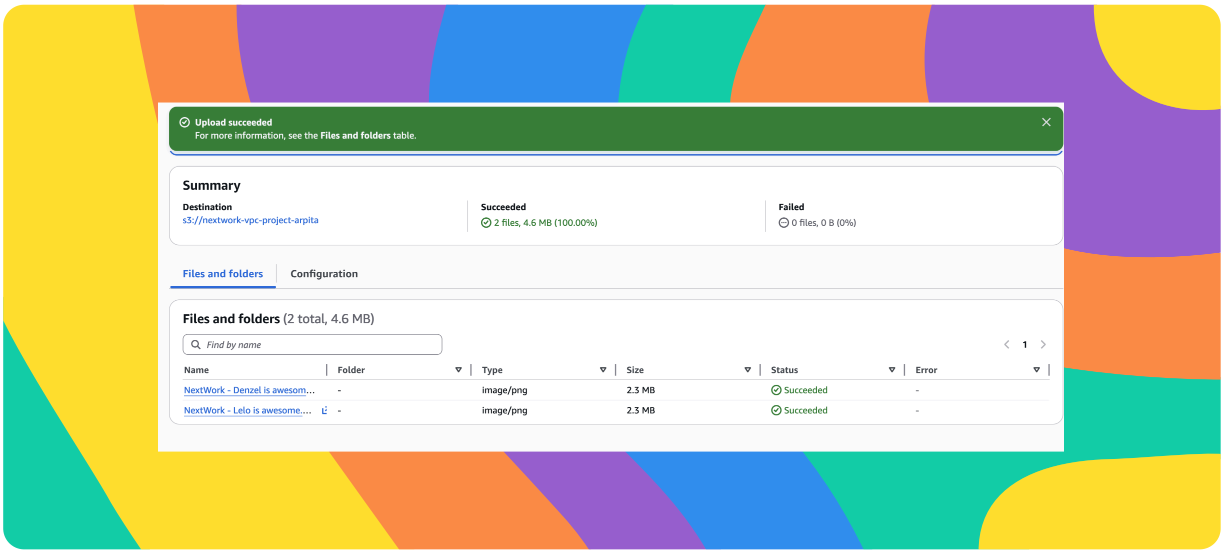
Architecture set up

I started my project by launching a VPC and EC2 instance.

I also set up a S3 bucket and uploaded two files NextWork - Denzel is awesome.png and NextWork - Lelo is awesome.png from local computer.



Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured my access key ID, secret access key, region name and default output format.

Access keys are credentials that allow programmatic access to AWS services via the CLI, SDKs, or API.

Secret access keys are the private half of an AWS access key pair, used to securely sign programmatic requests to AWS services.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use IAM roles.

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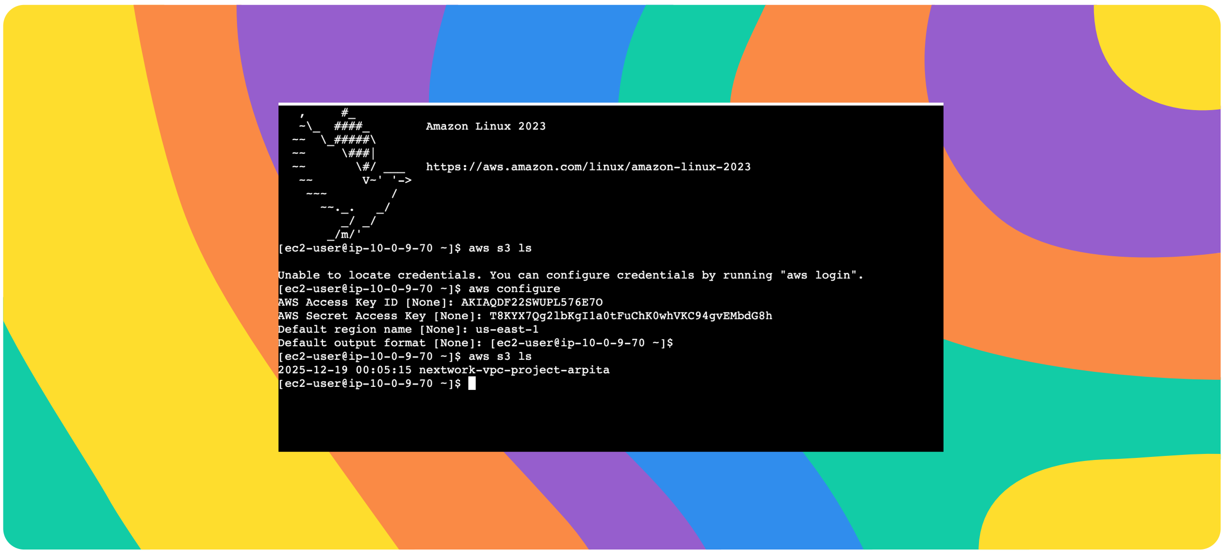
Connecting to my S3 bucket

The command I ran was aws S3 ls. This command is used to list all content present in S3 bucket.

The terminal responded with aws s3 ls. This indicated that the access keys I set up were valid and had sufficient permissions, because the AWS CLI was able to successfully authenticate and list the S3 bucket.



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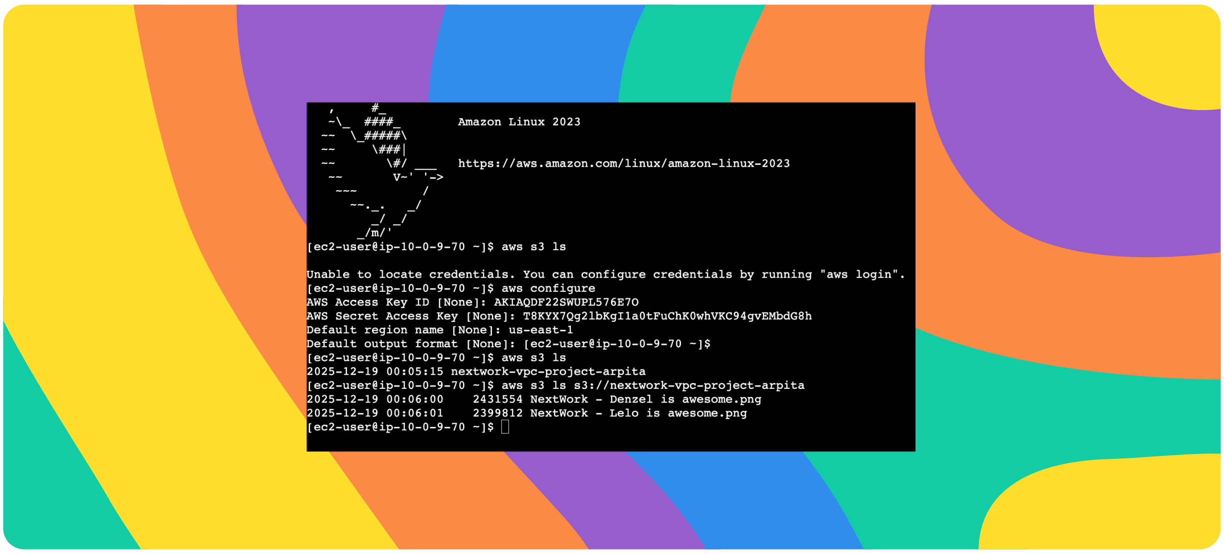
Connecting to my S3 bucket

I also tested the command aws s3 ls s3://nextwork-vpc-project-arpita which returned the files in s3 bucket.



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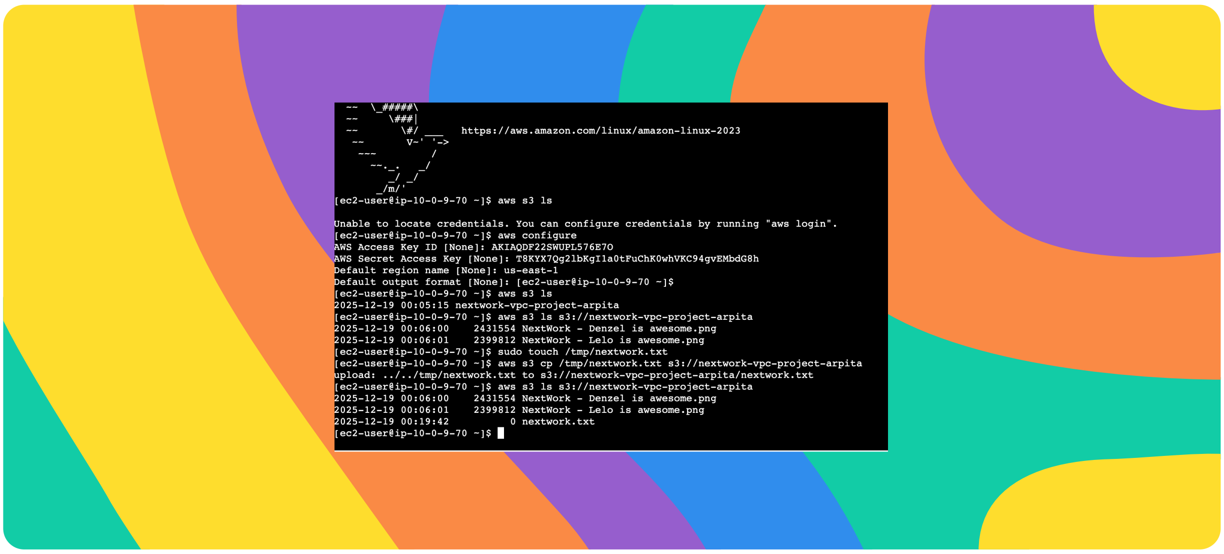
Uploading objects to S3

To upload a new file to my bucket, I first ran the command sudo touch /tmp/network.txt. This command creates an empty file named network.txt in the /tmp directory, which I then uploaded to the S3 bucket using the AWS CLI.

The second command I ran was aws s3 cp /tmp/network.txt s3://nextwork-vpc-project-arpita/network.txt. This command will upload the local file network.txt from the /tmp directory to the specified S3 bucket, creating the object in Amazon S3.

The third command I ran was aws s3 ls s3://nextwork-vpc-project-arpita, which validated that the file had been successfully uploaded to the S3 bucket by showing network.txt in the bucket’s contents.



In the second part of my project...

Step 5 - Set up a Gateway

In this step, I will set up a way for my VPC and S3 to communicate direclty because I want to create a VPC endpoint.

Step 6 - Bucket policies

In this step, I’m going to update my S3 bucket policy so that it blocks all access to the bucket except requests that come through my VPC’s S3 Gateway endpoint. This will let me verify that my EC2 instance is truly accessing S3 privately through the endpoint—because if the endpoint is working correctly, the instance will still be able to interact with the bucket even though all public internet access is denied.

Step 7 - Update route tables

In this step, I will test my VPC setup because I want to troubleshoot a connectivity issue.

Step 8 - Validate endpoint conection

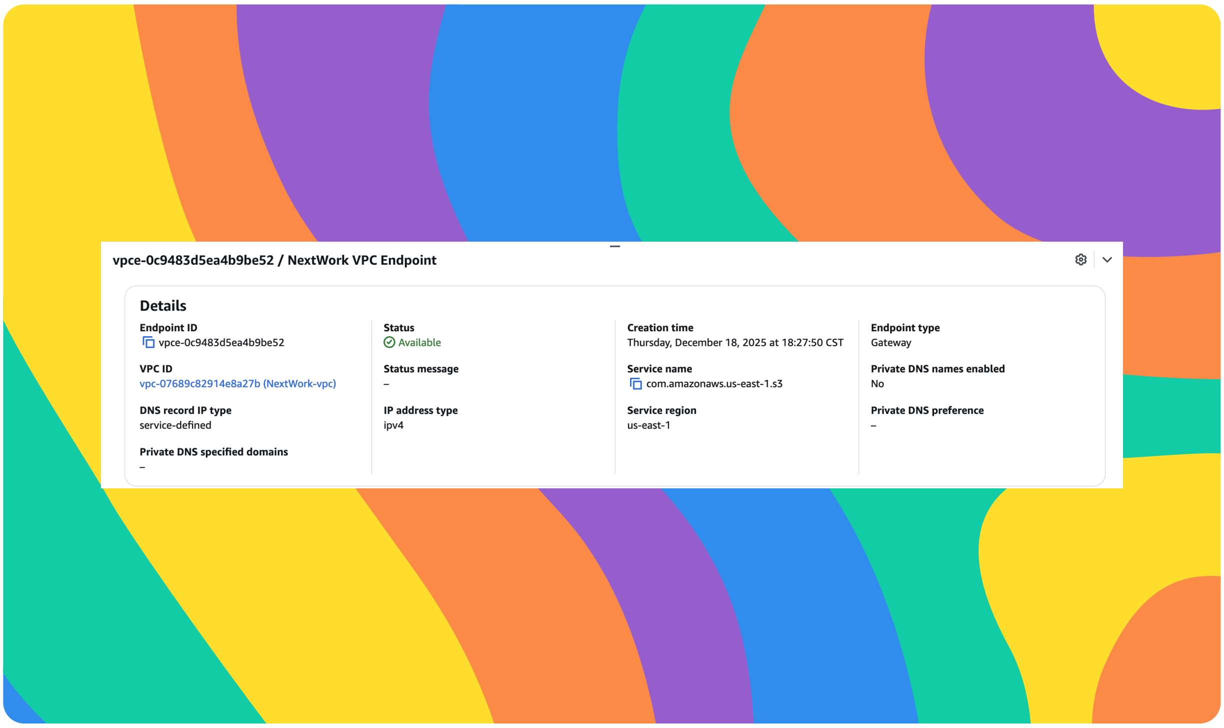
In this step, I will restrict my VPC access to my AWS enviroment because I want to test my VPC endpoint setup.

Setting up a Gateway

I set up an S3 Gateway, which is a VPC endpoint that allows resources within my VPC to privately access Amazon S3 without using the public internet, improving security and reducing data transfer costs.

What are endpoints?

An endpoint is a specific network address (URL or IP) that serves as an entry point to a service, allowing clients or applications to connect to and communicate with that service.



Bucket policies

A bucket policy is a resource-based policy attached to an Amazon S3 bucket that defines who can access the bucket and what actions they are allowed or denied, using rules based on principals, actions, resources, and conditions.

My bucket policy will deny all access to the S3 bucket by default and only allow requests that originate from my VPC’s S3 Gateway endpoint, ensuring the bucket can be accessed privately and not through the public internet.

```json
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": [
        "arn:aws:s3:::arn:aws:s3:::nextwork-vpc-project-arpita",
        "arn:aws:s3:::arn:aws:s3:::nextwork-vpc-project-arpita/*"
      ],
      "Condition": {
        "StringNotEquals": {
          "aws:sourceVpce": "vpce-0c9483d5ea4b9be52"
        }
      }
    }
  ]
}
```

Bucket policies

Right after saving my bucket policy, my S3 bucket page showed 'denied access' warnings. This was because this was because my new bucket policy denies all S3 actions unless the request comes through my VPC endpoint. Since the AWS Management Console isn’t accessing the bucket via that endpoint, the console’s requests don’t meet the policy condition and are explicitly denied, so the console shows “Access denied” warnings.

I also had to update my route table because If my route table doesn't have a route that directs traffic bound for S3 to my VPC endpoint, traffic from my EC2 instance is actually trying to get to my S3 bucket through the public internet instead.

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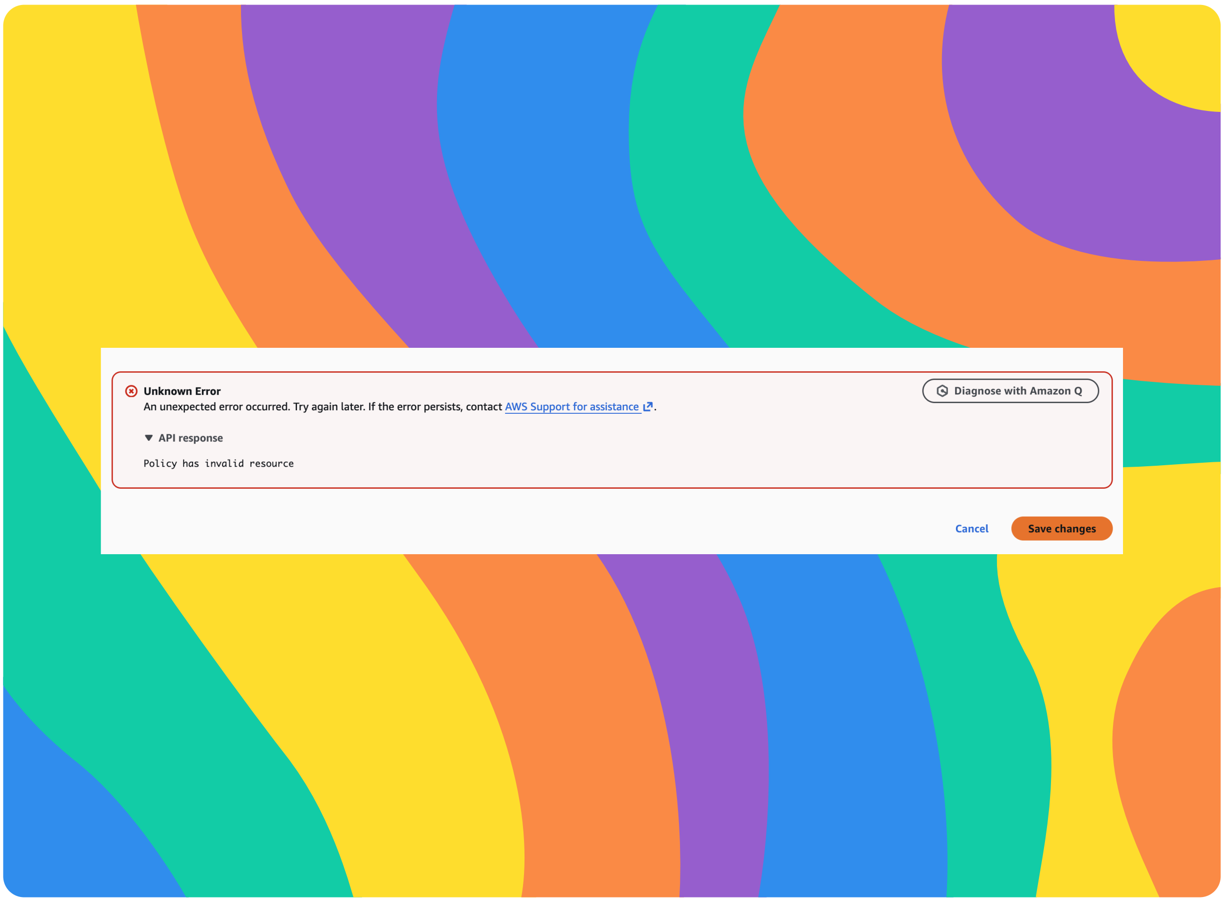

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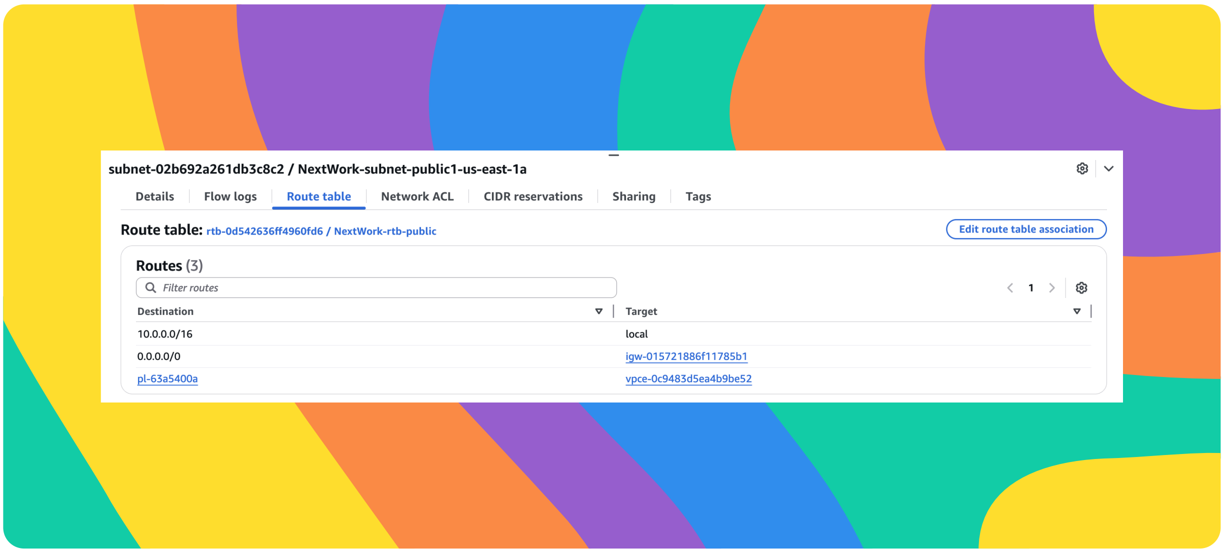
Route table updates

To update my route table, I did the following steps: I went to my VPC page and went to endpoint, Select a route table, Select manage route table, Select the checkbox next to my public route table and Select Modify route tables.

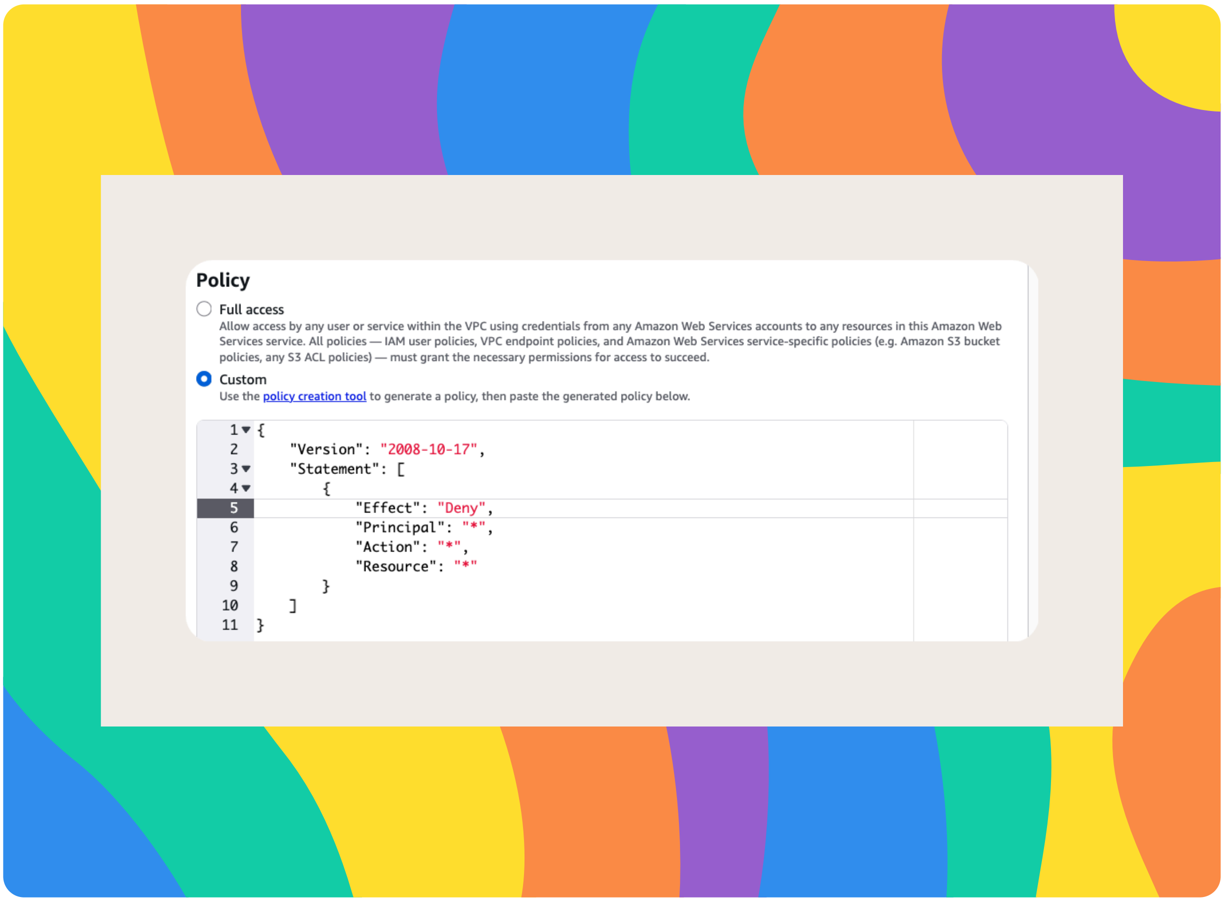
After updating my public subnet's route table, my terminal could return the files in the S3 bucket.



Endpoint policies

An endpoint policy is an IAM-style policy attached to a VPC endpoint that controls which AWS services and resources can be accessed through that endpoint, adding an extra layer of access control beyond IAM and resource policies.

I updated my endpoint's policy by changing the line "Effect": "Allow" to "Effect": "Deny". I could see the effect of this right away, because after enterning the command aws s3 ls s3://nextwork-vpc-project-arpita in EC3=2 treminal I was getting an error message saying access denied.



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